Challenge 1 - AI Security

Phase 0: Lab Setup & Environment Initialization

Before you can hack or defend, set up your local lab environment. The AI models are simulated in pure Python and NumPy to avoid heavy machine learning frameworks.

Step 1: Install Dependencies

Open your terminal and install the required Python libraries:

```
pip install numpy flask opencv-python
```

Step 2: Generate the Mock AI Models

Save the following code as `generate_models.py` in your project folder and run it. This creates the ./models/ directory and generates the `.pkl` files you need.

```python
# generate_models.py
import pickle
import numpy as np
import os

os.makedirs("./models", exist_ok=True)

class MockFaceRecognitionModel:
    def predict(self, image_tensor):
        if not isinstance(image_tensor, np.ndarray):
            raise TypeError(f"Model Error: Expected numpy.ndarray, got
{type(image_tensor)}")
        if image_tensor.shape != (1, 224, 224, 3):
            raise ValueError(f"Model Error: Invalid input shape. Expected
(1, 224, 224, 3), got {image_tensor.shape}")
        return {"label": "authorized", "confidence": 0.98}

class MockObjectDetectionModel:
    def predict(self, image_tensor):
        if not isinstance(image_tensor, np.ndarray):
            raise TypeError(f"Model Error: Expected numpy.ndarray, got
{type(image_tensor)}")
        if image_tensor.shape != (1, 224, 224, 3):
            raise ValueError(f"Model Error: Invalid input shape. Expected
(1, 224, 224, 3), got {image_tensor.shape}")
        return [
            {"label": "person", "confidence": 0.95},
            {"label": "knife", "confidence": 0.88},
            {"label": "backpack", "confidence": 0.75}
        ]

face_model = MockFaceRecognitionModel()
object_model = MockObjectDetectionModel()
```

```
with open("./models/face_recognition_model.pkl", "wb") as f:
    pickle.dump(face_model, f)
with open("./models/object_detection_model.pkl", "wb") as f:
    pickle.dump(object_model, f)

print("Mock models generated successfully in ./models/")
```

Run it via terminal: `python generate_models.py`

Step 3: Prepare Local Directories & Update Paths

The original code uses absolute root paths (`/models/...` and `/var/log/...`) which will cause "Permission Denied" errors on your local machine.

1. Create a local log folder: `mkdir ./logs`
2. Open `lab_access_api.py` and **change the paths** to use local relative paths:
    - Change `"/models/face_recognition_model.pkl"` to `"./models/face_recognition_model.pkl"`
    - Change `"/models/object_detection_model.pkl"` to `"./models/object_detection_model.pkl"`
    - Change `"/var/log/lab_access/images/"` to `"./logs/"`